



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 5

1. Case Study Title: To study the functioning of arrays and loops in JavaScript.

Create and manipulate arrays: push, pop, shift, unshift, splice, slice.

Use array methods: map(), filter(), reduce(), and forEach().

Create an array of numbers and find the maximum and minimum values.

2. Case Study Program Code:

```
//Sonal Nakhate 24070521199
```

```
// 1. Create an Array
```

```
let fruits = ["apple", "banana", "cherry"];
```

```
// push() - Adds an element to the END
```

```
fruits.push("mango");
```

```
// Result: ["apple", "banana", "cherry", "mango"]
```

```
// pop() - Removes the LAST element
```

```
let lastFruit = fruits.pop();
```

```
// Result: ["apple", "banana", "cherry"] | Removed: "mango"
```

```
// unshift() - Adds an element to the BEGINNING
```

```
fruits.unshift("strawberry");
```

```
// Result: ["strawberry", "apple", "banana", "cherry"]
```

```
// shift() - Removes the FIRST element
```



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```
let firstFruit = fruits.shift();
// Result: ["apple", "banana", "cherry"] | Removed: "strawberry"

// splice(startIndex, deleteCount, ...itemsToAdd) - Modifies original
array (Mutates)
// Example: Remove 1 element at index 1 and insert "grape"
fruits.splice(1, 1, "grape");
// Result: ["apple", "grape", "cherry"]

// slice(startIndex, endIndex) - Extracts a portion WITHOUT modifying
original (Non-mutating)
// Note: endIndex is non-inclusive
let subArray = fruits.slice(0, 2);
// subArray: ["apple", "grape"]
// fruits stays: ["apple", "grape", "cherry"]

const numbers = [1, 2, 3, 4, 5, 6];

// 1. forEach() - Executes a function once for each array element (returns
nothing)
numbers.forEach((num) => {
  console.log(`Item: ${num}`);
});

// 2. map() - Creates a NEW array by transforming every element
const doubled = numbers.map((num) => num * 2);
// Result: [2, 4, 6, 8, 10, 12]

// 3. filter() - Creates a NEW array with elements that satisfy a
condition
const evens = numbers.filter((num) => num % 2 === 0);
// Result: [2, 4, 6]

// 4. reduce() - Accumulates array elements down to a SINGLE value
```



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```
// Syntax: array.reduce((accumulator, currentValue) => ..., initialValue)
const sum = numbers.reduce((acc, curr) => acc + curr, 0);
// Result: 21
```

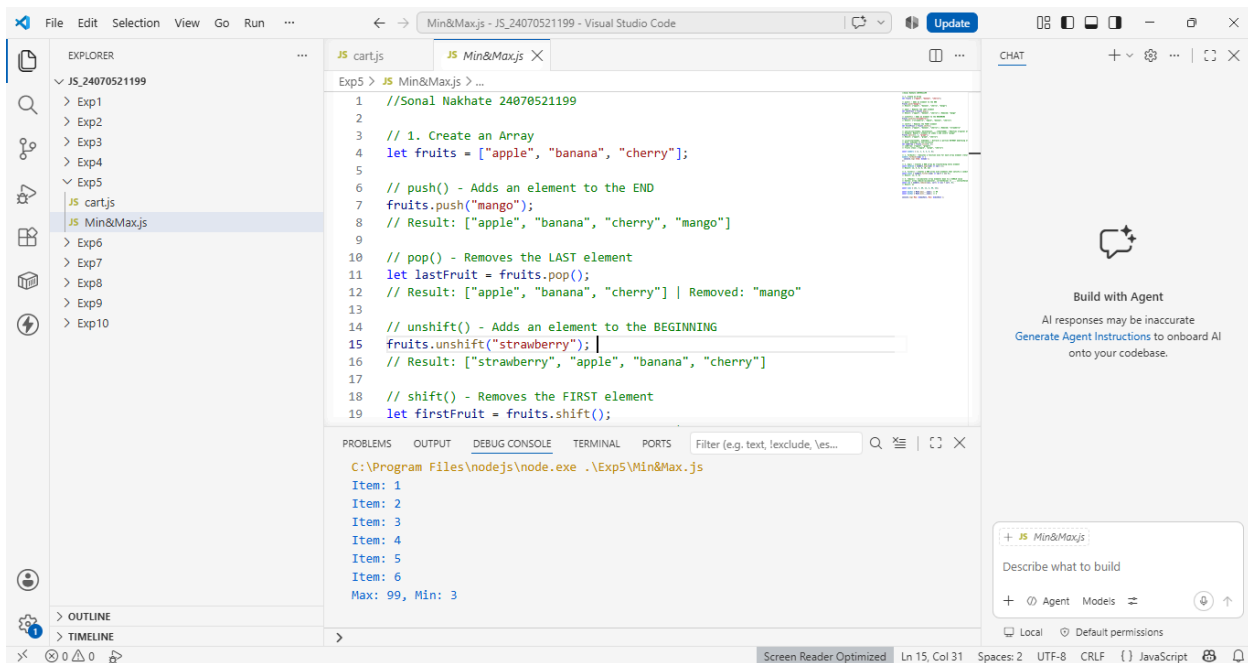
```
const nums = [42, 7, 89, 12, 3, 99, 23];
```

```
const maxVal = Math.max(...nums); // 99
```

```
const minVal = Math.min(...nums); // 3
```

```
console.log(`Max: ${maxVal}, Min: ${minVal}`);
```

3. Output:



4. Result/Conclusion: This experiment demonstrated the core mechanics of JavaScript arrays and iterative processing.



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Highlighting how different methods handle data manipulation, transformation, and evaluation. Modifying arrays using `push()`, `pop()`, `shift()`, `unshift()`, `splice()`, and `slice()` established a practical understanding of element positioning and the distinction between mutating original data versus returning new copies. Furthermore, utilizing functional array methods like `forEach()`, `map()`, `filter()`, and `reduce()` illustrated a clean, declarative alternative to traditional looping for transforming and aggregating data. Finally, calculating the maximum and minimum values of a number array using `Math.max()/Math.min()` with the spread operator, `reduce()`, and classic `for` loops demonstrated how algorithmic logic and modern language features can solve fundamental computational problems efficiently.